MILITARY SPECIFICATION

TRANSDUCER, MOTIONAL PICKUP TR-299/G( )

This amendment forms a part of Military Specification MIL-T-38531(17), dated 15 March 1976, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Following the source paragraph at the end of 2.1, add the following new paragraph:

"2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United Technical Publications

Electronic Engineer's Master (EEM)

(Application for copies of EEM should be addressed to United Technical Publications, Inc., 645 Stewart Ave., Garden City, N.Y. 11530.)"

PAGE 3

3.3.1.3.2: Delete and substitute:

"3.3.1.3.2 Operational sensitivity. The sample mean value of detection probability of a set of three test transducers when tested in accordance with 4.8.4 shall not be more than 5 percent below mean value of a set of three reference transducers tested concurrently. The sample mean value of false alarm rate for the test set of transducers shall not be more than 5 percent above mean value of the reference set tested concurrently. The false alarm rate will be computed as the number of false alarms (defined as the total number of alarms minus the number of nuisance alarms recorded during the test) divided by the total accumulated test time. A nuisance alarm is defined as an alarm signal generated by a transducer when a non-human intruder (e.g., a small animal) is within the sensitivity zone of the transducer."

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PAGE 5

* 3.5a: Delete the second sentence.

PAGE 9

* 4.8.1a.(2): Delete and substitute:

"(2) High temperature: Method 501.1, Procedure II, Step 7 raise internal chamber temperature to 71°C and maintain until temperature stabilization of the transducer is reached. In Step 8 the transducer shall be operated for a minimum of one hour."

PAGE 11

4.8.1c.(2)(g): Delete and substitute:

"(g) At the end of observation period, raise platform out of water, release vacuum, and open tank. Record any failures identifying the unit that failed."

PAGE 12

4.8.1d: Delete and substitute:

"d Shock: Method 516.2 Procedure II."

PAGE 13

4.8.3: Delete and substitute:

"4.8.3 Core permeability test. This test shall be conducted on the stranded transducer core after wrapping with insulating tape but before solenoid windings are added. The instrumentation is shown in Figure 3. The relative permeability of the core at 50 Hz ± 1Hz shall meet the requirements of 3.3.1.3.1."

PAGE 14

4.8.3.1 and subparagraphs a thru h: Delete and substitute:

"4.8.3.1 Test procedures.

a. Connect the equipment as shown in Figure 3 without the TR-299/G( ) through the cylinder.

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b. Measure the cross-sectional area of the cylinder (coil form), A_s, and the cross-sectional area of the TR-299/G( ) core material minus the insulating tape, A_c.

c. Set the signal generator for 50 $\pm$1 Hz.

d. Increase the size of the signal from the signal generator until the voltmeter registers a deflection between 0.5 and 1 volt, Measure the secondary voltage, E_s.

e. Compute what the secondary voltage E_s' should be if the TR-299/G( ) core material was through the center of the cylinder for a permeability, μ_c, of 1000. Use the following equation to compute E_s'.

$$E_s' = \frac{E_s A_c \mu_c + E_s A_s}{A_s}$$

f. Set the voltmeter to close the relay (switch) which in turn triggers the buzzer (alarm) when the secondary voltage with the transducer core through the cylinder center drops below E_s'.

g. Feed the transducer core through the cylinder center at a rate not greater than 2 meters/second."

PAGE 15

4.8.3.2: Delete and substitute:

"4.8.3.2 Accept-reject criteria. A defective transducer core shall be identified by failure to have a measured E_s' below the minimum E_s' as computd by 4.8.3.1.e."

PAGE 21

* Temperature block: Delete references to "163" and "325".

PAGE 24

Figure 3: Delete and substitute.

PAGE 25

Figure 5: Change the minimum typical distance between transducers from "1.52 m" to "3.04 m".

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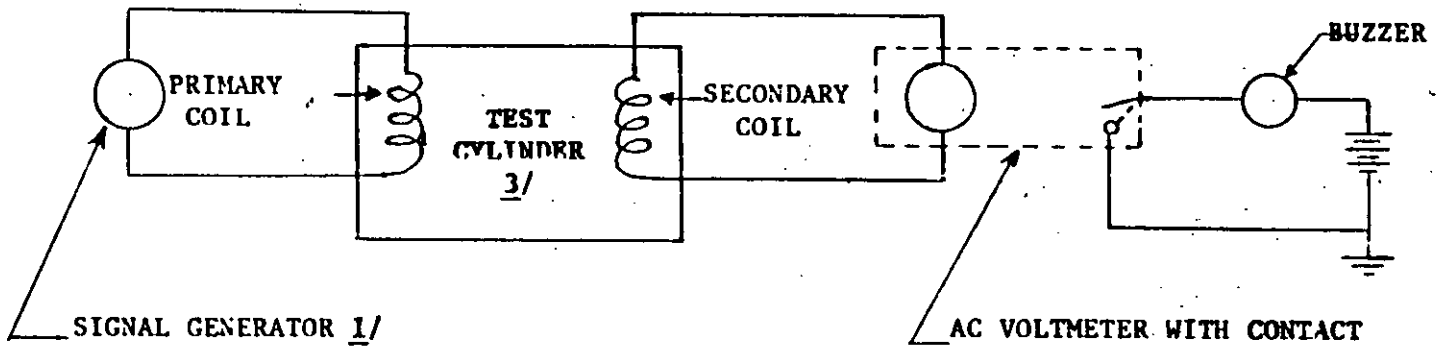
NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 6350-F102)

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1/ General Radio Type 1162A or equivalent. Accuracy: ±1%

2/ Listed in EEM under heading of "Relays, Meters".

3/ Test cylinder formed from a plexiglass cylinder (or any other non-magnetic/non-conducting cylinder) with a minimum length to diameter ratio of 2. Wind a primary coil tightly around the entire length of the cylinder (coil form). On top of the primary coil wind the secondary coil with the same number of turns. The number of turns is arbitrary, but the number for each coil must be the same.

FIGURE 3. CORE PERMEABILITY TEST

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